A Few Notes on Vector Calculus

David Meyer

dmm@1-4-5.net

Last update: March 22, 2022

1 Introduction

TBD

2 Notation

Some of the notation we'll use is described in this section.

2.1 Vectors

Some authors use bold font to denote a vector, e.g. $\mathbf{u}$. Adding a "hat" denotes a unit vector. That is, for a vector $\mathbf{u}$, $\hat{\mathbf{u}}$ is defined to be

$$\hat{\mathbf{u}} := \frac{\mathbf{u}}{\|\mathbf{u}\|}$$

In words: $\hat{\mathbf{u}}$ is a vector of unit length in the $\mathbf{u}$ direction.

Similarly, $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{i}}$ are unit vectors in $\mathbb{R}^3$ in the x, y, and z directions respectively. Note that $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are the canonical basis vectors for $\mathbb{R}^3$ [3] and have column vector format

$$\hat{\mathbf{i}} = \begin{bmatrix}1\\0\\0\end{bmatrix},\ \hat{\mathbf{j}} = \begin{bmatrix}0\\1\\0\end{bmatrix},\ \hat{\mathbf{k}} = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

I have also seen $\mathbf{e}_i$ used to represent the ith basis vector in $\mathbb{R}^n$. So $\mathbf{e}_i$ is a vector with a one in the ith position and a zero in each of the other $n-1$ positions. In $\mathbb{R}^3$ this means that $\mathbf{e}_1 = \hat{\mathbf{i}}$, $\mathbf{e}_2 = \hat{\mathbf{j}}$, and $\mathbf{e}_3 = \hat{\mathbf{k}}$.

In general, the standard basis (which is sometimes called the computational basis) for the n-dimensional Euclidean space consists of the ordered set of n distinct vectors

$$\{\mathbf{e}_i : 1 \leq i \leq n\}$$

where $\mathbf{e}_i$ is the ith basis vector, that is, it has a one in the ith coordinate (position) and zeros everywhere else.[1] The $\mathbf{e}_i$ have column vector format

[1] This is also called a "one-hot" encoding in machine learning parlance, where the i's might be the classes in a classification problem.

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \ldots, \mathbf{e}_{n-1} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix}, \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

Using these definitions we can define the parametric form of some curve C in $\mathbb{R}^3$, $\mathbf{r}(t)$, as follows:

$$\mathbf{r}(t) = g(t)\hat{\mathbf{i}} + h(t)\hat{\mathbf{j}} + k(t)\hat{\mathbf{k}}$$

where $t \in [a, b]$ and $g(t)$, $h(t)$, and $k(t)$ are scalar functions[2].

Another common notation for vectors is $\vec{r}(t)$, where $\vec{r}(t)$ might be defined as follows:

$$\vec{r}(t) = g(t)\hat{i} + h(t)\hat{j} + k(t)\hat{k}$$

In this document I'll use these notations interchangeably.

3 Definitions and a Few Theorems

Definition 3.1. Vector Notation:

Here we will use $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ to represent a column or row vector in some n-dimensional space (usually $\mathbb{R}$ or $\mathbb{C}$). If $\mathbf{a}$ is a column vector then in matrix format

$$\mathbf{a} = \underbrace{\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}}_{n \times 1}$$

Alternatively, if $\mathbf{a}$ is a row vector then in matrix format

$$\mathbf{a} = \underbrace{\begin{bmatrix} a_1, a_2, \ldots, a_n \end{bmatrix}}_{1 \times n}$$

The *transpose* of a vector $\mathbf{a}$, $\mathbf{a}^\intercal$, is defined as follows. If $\mathbf{a}$ is a column vector then

$$\mathbf{a}^\intercal = \begin{bmatrix} a_1, a_2, \ldots, a_n \end{bmatrix}$$

Alternatively, if $\mathbf{a}$ is a row vector then

$$\mathbf{a}^\intercal = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

[2] A scalar function is a function f such that $f : \mathbb{R}^n \to \mathbb{R}$.

Definition 3.2. Dot Product: $\mathbf{a} \cdot \mathbf{b}$

The dot product (aka scalar product or inner product) of two n-dimensional vectors $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$, denoted $\mathbf{a} \cdot \mathbf{b}$, is defined to be the scalar value

$$\mathbf{a} \cdot \mathbf{b} := \sum_{j=1}^{n} a_j b_j \tag{1}$$

Another common notation for the dot product of $\mathbf{a}$ and $\mathbf{b}$ is $\langle \mathbf{a}, \mathbf{b} \rangle$. That is, $\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a} \cdot \mathbf{b}$.

If $\mathbf{a}$ and $\mathbf{b}$ are column vectors then their dot product $\mathbf{a} \cdot \mathbf{b}$ can also be written as the matrix product $\mathbf{a}^\intercal \mathbf{b}$. Similarly, if $\mathbf{a}$ and $\mathbf{b}$ are row vectors then their dot product can be written as the matrix product $\mathbf{a}\mathbf{b}^\intercal$. For example, the dot product of two $n \times 1$ column vectors $\mathbf{a}$ and $\mathbf{b}$ is equivalent to the following matrix multiplication:

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{a}^\intercal \mathbf{b} \to \begin{bmatrix} a_1, a_2, \ldots, a_n \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \to \sum_{j=1}^{n} a_j b_j \to c$$

where $c \in \mathbb{R}$.

According to the Pythagorean theorem, the length of a vector $\mathbf{a} = (a_1, a_2, a_3) \in \mathbb{R}^3$, denoted $\|\mathbf{a}\|$, equals $\sqrt{a_1^2 + a_2^2 + a_3^2}$. The next definition is a generalization of the notion of length to vectors in $\mathbb{R}^n$.

Definition 3.3. Euclidean Norm: $\|\mathbf{x}\|$

The Euclidean norm of a vector

$$\mathbf{x} = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$$

is defined to be

$$\|\mathbf{x}\| := \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = \sqrt{\sum_{j=1}^{n} x_j^2} \tag{2}$$

Definition 3.4. p-Norm: $\|\mathbf{x}\|_p$

The p-norm of a vector

$$\mathbf{x} = (x_1, x_2 \ldots, x_n)$$

is defined to be

$$\|\mathbf{x}\|_p := \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$$

for $p \geq 1$ and $p \in \mathbb{R}$.

For $p = 1$, we get the taxicab norm, for $p = 2$ we get the Euclidean norm, and as p approaches ∞ the p-norm approaches the infinity norm or maximum norm: $\|\mathbf{x}\|_\infty := \max_i |x_i|$.

Next, assume that $\mathbf{x}$ and $\mathbf{y}$ are two linearly independent vectors in $\mathbb{R}^3$ and that M is the plane spanned by them. Then these two vectors generate a triangle in M with sides of length $\|\mathbf{x}\|$, $\|\mathbf{y}\|$, and $\|\mathbf{x} - \mathbf{y}\|$. If $\theta \in \{0, \pi\}$ is the angle between the vectors $\mathbf{x}$ and $\mathbf{y}$ in M, then the Law of Cosines [8] gives us

$$\|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\cos(\theta)\|\mathbf{x}\|\|\mathbf{y}\|$$

We can also see that

$$\begin{aligned}
\|\mathbf{x} - \mathbf{y}\|^2 &= \langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle && \text{\# by Equation (2)} \\
&= \sum_{j=1}^{n} (x_j - y_j)^2 && \text{\# by Equation (2)} \\
&= \sum_{j=1}^{n} (x_i^2 - 2x_j y_j + y_j^2) && \text{\# expand previous line} \\
&= \sum_{j=1}^{n} x_j^2 - 2\sum_{j=1}^{n} x_j y_j + \sum_{j=1}^{n} y_j && \text{\# sum is a linear operator} \\
&= \|\mathbf{x}\|^2 + -2\langle \mathbf{x}, \mathbf{y} \rangle + \|\mathbf{y}\|^2 && \text{\# by Equation (2)} \\
&= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\langle \mathbf{x}, \mathbf{y} \rangle && \text{\# rearrange}
\end{aligned}$$

Combining these expressions for $\|\mathbf{x} - \mathbf{y}\|^2$ we get

$$\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\cos(\theta)\|\mathbf{x}\|\|\mathbf{y}\| = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\langle \mathbf{x}, \mathbf{y} \rangle$$

Subtracting $\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$ from both sides, dividing by -2, and rearranging gives us

$$\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\|\|\mathbf{y}\| \cos\theta \tag{3}$$

Definition 3.5. Orthogonal Vectors: $\langle \mathbf{x}, \mathbf{y} \rangle = 0$

We say that $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are orthogonal if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$, since

$$\begin{aligned}
\langle \mathbf{x}, \mathbf{y} \rangle &= \|\mathbf{x}\|\|\mathbf{y}\| \cos\theta && \text{\# by Equation (3)} \\
&\Rightarrow \langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\|\|\mathbf{y}\| \cos\tfrac{\pi}{2} && \text{\# } \mathbf{x} \text{ and } \mathbf{y} \text{ orthogonal} \Rightarrow \theta = \tfrac{\pi}{2} \\
&\Rightarrow \langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\|\|\mathbf{y}\|\, 0 && \text{\# } \cos\tfrac{\pi}{2} = 0 \\
&\Rightarrow \langle \mathbf{x}, \mathbf{y} \rangle = 0 && \text{\# } \|\mathbf{x}\|\|\mathbf{y}\|\, 0 = 0
\end{aligned}$$

Definition 3.6. Orthogonal Projection:

Given two linearly independent vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, we want to find the orthogonal projection of $\mathbf{a}$ onto the line generated by $\mathbf{b}$. To analyze this situation, denote by M the plane generated by $\mathbf{a}$ and $\mathbf{b}$ and consider an orthonormal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ of M consisting of $\mathbf{v}_1 = \frac{\mathbf{b}}{\|\mathbf{b}\|}$ ($\mathbf{v}_1 = \hat{\mathbf{b}}$) and a unit vector $\mathbf{v}_2 \in M$ orthogonal to $\mathbf{v}_1$. This situation is shown in Figure 1.

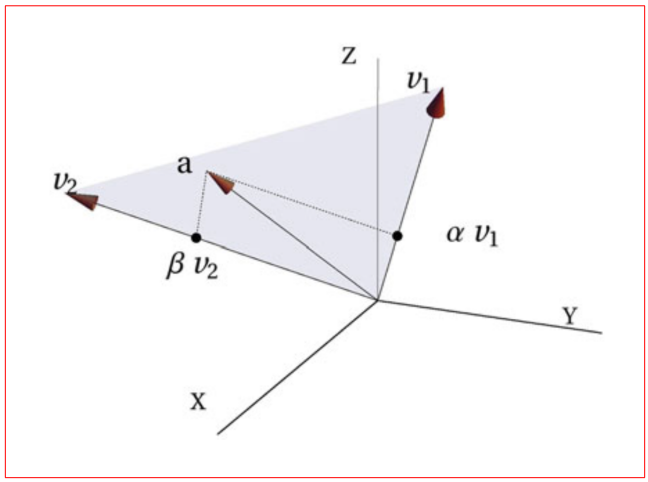

Figure 1: Orthogonal Projection

Since $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis of M there are scalars α and β such that

$$\mathbf{a} = \alpha \mathbf{v}_1 + \beta \mathbf{v}_2 \tag{4}$$

Here we want to solve for α. One way to do that is to take the inner product of $\mathbf{b}$ and Equation (4), since

$$\begin{aligned}
\langle \mathbf{a}, \mathbf{b} \rangle &= \langle \mathbf{b}, \mathbf{a} \rangle && \text{\# by Equation (1)} \\
&= \langle \mathbf{b}, (\alpha \mathbf{v}_1 + \beta \mathbf{v}_2) \rangle && \text{\# by Equation (4)} \\
&= \langle \mathbf{b}, \alpha \mathbf{v}_1 \rangle + \langle \mathbf{b}, \beta \mathbf{v}_2 \rangle && \text{\# dot product is distributive over vector addition [13]} \\
&= \alpha \langle \mathbf{b}, \mathbf{v}_1 \rangle + \beta \langle \mathbf{b}, \mathbf{v}_2 \rangle && \text{\# by the scalar product rule: } \langle c_1 \mathbf{v}_1, c_2 \mathbf{v}_2 \rangle = c_1 c_2 \langle \mathbf{v}_1, \mathbf{v}_2 \rangle \text{ [13]} \\
&= \alpha \langle \mathbf{v}_1, \mathbf{b} \rangle + \beta\, 0 && \text{\# since } \langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 0 \text{ and } \mathbf{v}_1 = \hat{\mathbf{b}} \Rightarrow \langle \hat{\mathbf{b}}, \mathbf{v}_2 \rangle = \tfrac{1}{\|\mathbf{b}\|} \langle \mathbf{b}, \mathbf{v}_2 \rangle = 0 \Rightarrow \langle \mathbf{b}, \mathbf{v}_2 \rangle = 0 \\
&= \alpha \langle \mathbf{v}_1, \mathbf{v_1} \|\mathbf{b}\| \rangle && \text{\# since } \beta\, 0 = 0 \text{ and } \mathbf{v}_1 = \tfrac{\mathbf{b}}{\|\mathbf{b}\|} \text{ so } \mathbf{b} = \mathbf{v_1} \|\mathbf{b}\| \\
&= \alpha \|\mathbf{b}\| \langle \mathbf{v}_1, \mathbf{v}_1 \rangle && \text{\# by the scalar product rule for dot products} \\
&= \alpha \|\mathbf{b}\| && \text{\# since } \mathbf{v}_1 = \tfrac{\mathbf{b}}{\|\mathbf{b}\|} \Rightarrow \|\mathbf{v}_1\| = 1 \text{ and so } \langle \mathbf{v}_1, \mathbf{v_1} \rangle = \|\mathbf{v}_1\|^2 = 1^2 = 1
\end{aligned}$$

So we see that

$$\langle \mathbf{a}, \mathbf{b} \rangle = \alpha \|\mathbf{b}\|$$

and so

$$\alpha = \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{b}\|} = \left\langle \mathbf{a}, \frac{\mathbf{b}}{\|\mathbf{b}\|} \right\rangle = \langle \mathbf{a}, \hat{\mathbf{b}} \rangle$$

α represents the component of the vector $\mathbf{a}$ parallel to ("in the direction of") the vector $\hat{\mathbf{b}}$. More generally, since $\hat{\mathbf{b}}$ is the unit vector in the $\mathbf{b}$ direction, $\langle \mathbf{a}, \mathbf{b} \rangle$ is also the component of $\mathbf{a}$ in the $\mathbf{b}$ direction.

Definition 3.7. The Cross Product of Two Vectors: $\mathbf{a} \times \mathbf{b}$

The cross product is a binary operation on two vectors in a three-dimensional oriented Euclidean vector space [4]. Specifically, given two linearly independent vectors $\mathbf{a}$ and $\mathbf{b}$, the cross product of $\mathbf{a}$ and $\mathbf{b}$, denoted $\mathbf{a} \times \mathbf{b}$, is defined to be a vector with the following three properties:

- $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ (Figure 2)
- $\mathbf{a} \times \mathbf{b}$ has direction given by the right-hand rule (Figure 3)
- $\mathbf{a} \times \mathbf{b}$ has magnitude equal to the area of the parallelogram that the vectors span (Figure 2)

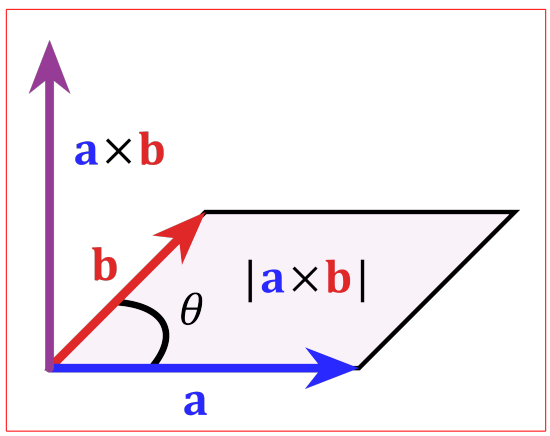

Figure 2: The parallelogram formed by $\mathbf{a} \times \mathbf{b}$ [12]

One definition of the cross product is

$$\mathbf{a} \times \mathbf{b} = \|\mathbf{a}\| \, \|\mathbf{b}\| \sin(\theta) \, \mathbf{n} \tag{5}$$

where

- θ is the angle between $\mathbf{a}$ and $\mathbf{b}$ in the plane containing them, so $0 \leq \theta \leq \pi$

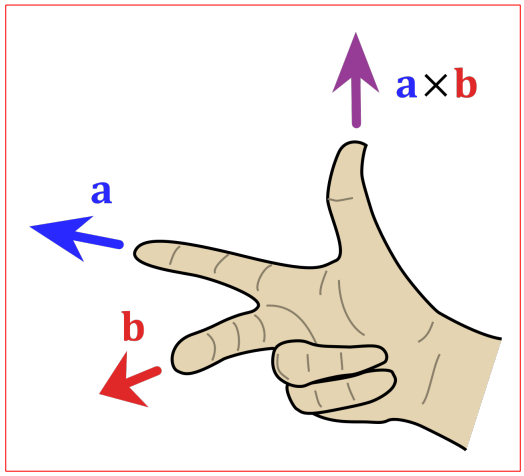

Figure 3: The right hand rule for vectors **a** and **b** [15]

- $\|\mathbf{a}\|$ and $\|\mathbf{b}\|$ are the magnitudes of vectors **a** and **b**
- **n** is a unit vector normal to the plane containing **a** and **b** in the direction given by the right-hand rule

If the cross product of two vectors is the zero vector (that is, $\mathbf{a} \times \mathbf{b} = \mathbf{0}$), then either one or both of the inputs is the zero vector ($\mathbf{a} = \mathbf{0}$ and/or $\mathbf{b} = \mathbf{0}$) or the vectors are parallel or anti-parallel ($\theta \in \{0, \pi\}$). In the second case ($\theta \in \{0, \pi\}$), since the angle between a vector and itself is zero, Equation (5) tells us that the cross product of a vector with itself is the zero vector: $\mathbf{a} \times \mathbf{a} = \|\mathbf{a}\| \, \|\mathbf{a}\| \sin(0) \, \mathbf{n} = \|\mathbf{a}\| \, \|\mathbf{a}\| \, 0 \, \mathbf{n} = \mathbf{0}$.

Note that by the right-hand rule, if the thumb points at you then the fingers curl in the anti-clockwise (counter-clockwise) direction. That is, when viewed from the top or z axis the system rotates in an anti-clockwise direction. This is one of the reasons we take the direction of a curve to be in the anti-clockwise direction when we consider, for example, line integrals in Section 5.

The cross product can also be expressed as the following determinant

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

which can be computed using Sarrus's rule [11] or cofactor expansion [2]. Using Sarrus's rule the cross product expands to

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= (a_2 b_3 \hat{\mathbf{i}} + a_3 b_1 \hat{\mathbf{j}} + a_1 b_2 \hat{\mathbf{k}}) - (a_3 b_2 \hat{\mathbf{i}} + a_1 b_3 \hat{\mathbf{j}} + a_2 b_1 \hat{\mathbf{k}}) \\ &= (a_2 b_3 - a_3 b_2)\hat{\mathbf{i}} + (a_3 b_1 - a_1 b_3)\hat{\mathbf{j}} + (a_1 b_2 - a_2 b_1)\hat{\mathbf{k}} \end{aligned}$$

Using cofactor expansion the 3×3 determinant can be expressed in terms of 2×2 determinants:

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \hat{\mathbf{i}} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \hat{\mathbf{j}} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \hat{\mathbf{k}} \\ &= (a_2 b_3 - a_3 b_2)\hat{\mathbf{i}} - (a_1 b_3 - a_3 b_1)\hat{\mathbf{j}} + (a_1 b_2 - a_2 b_1)\hat{\mathbf{k}} \end{aligned}$$

The cross product has the following additional properties [12]:

- it is anti-commutative: $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$
- it is distributive over addition: $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$
- it is compatible with scalar multiplication so that: $(c\,\mathbf{a}) \times \mathbf{b} = \mathbf{a} \times (c\,\mathbf{b}) = c\,(\mathbf{a} \times \mathbf{b})$
- it is not associative but satisfies the Jacobi identity [14]: $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{0}$

Theorem 3.1. Cauchy–Schwarz Inequality: $\left|\langle \mathbf{a}, \mathbf{b} \rangle\right| \leq \|\mathbf{a}\|\|\mathbf{b}\|$

Proof: The inequality is trivial if either $\mathbf{a}$ or $\mathbf{b}$ is zero, so assume that neither is. So without loss of generality let $\mathbf{x} = \dfrac{\mathbf{a}}{\|\mathbf{a}\|}$ and $\mathbf{y} = \dfrac{\mathbf{b}}{\|\mathbf{b}\|}$ and so $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$ ($\mathbf{x} = \hat{\mathbf{a}}$ and $\mathbf{y} = \hat{\mathbf{b}}$). Then

$$
\begin{aligned}
0 &\leq \|\mathbf{x} - \mathbf{y}\| && \text{\# } \|\mathbf{x} - \mathbf{y}\| \geq 0 \text{ for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n \text{ by Equation (2)} \\
&\leq \|\mathbf{x} - \mathbf{y}\|^2 && \text{\# } \|\mathbf{x} - \mathbf{y}\| \geq 0 \Rightarrow \|\mathbf{x} - \mathbf{y}\|^2 \geq 0 \\
&\leq \sum_{j=1}^{n} (x_j - y_j)^2 && \text{\# } \|\mathbf{x} - \mathbf{y}\|^2 = \sum_{j=1}^{n} (x_j - y_j)^2 \text{by Equation (2)} \\
&\leq \sum_{j=1}^{n} \left(x_j^2 - 2x_j y_j + y_j^2\right) && \text{\# expand previous line} \\
&\leq \sum_{j=1}^{n} x_j^2 - 2\sum_{j=1}^{n} x_j y_j + \sum_{j=1}^{n} y_j^2 && \text{\# sum is a linear operator} \\
&\leq \|\mathbf{x}\|^2 - 2\langle \mathbf{x}, \mathbf{y} \rangle + \|\mathbf{y}\|^2 && \text{\# } \|\mathbf{x}\|^2 = \sum_{j=1}^{n} x_j^2 \text{ and } \langle \mathbf{x}, \mathbf{y} \rangle = \sum_{j=1}^{n} x_j y_j \text{ by Equation (2)} \\
&\leq 1 - 2\langle \mathbf{x}, \mathbf{y} \rangle + 1 && \text{\# } \|\mathbf{x}\| = \|\mathbf{y}\| = 1 \Rightarrow \|\mathbf{x}\|^2 = \|\mathbf{y}\|^2 = 1 \\
&\leq 2 - 2\langle \mathbf{x}, \mathbf{y} \rangle && \text{\# } 1 + 1 = 2 \\
&\Rightarrow -2 \leq -2\langle \mathbf{x}, \mathbf{y} \rangle && \text{\# subtract 2 from both sides} \\
&\Rightarrow 1 \geq \langle \mathbf{x}, \mathbf{y} \rangle && \text{\# divide by -2, reversing inequality} \\
&\Rightarrow \langle \mathbf{x}, \mathbf{y} \rangle \leq 1 && \text{\# rearrange} \\
&\Rightarrow \left\langle \frac{\mathbf{a}}{\|\mathbf{a}\|}, \frac{\mathbf{b}}{\|\mathbf{b}\|} \right\rangle \leq 1 && \text{\# } \mathbf{x} = \frac{\mathbf{a}}{\|\mathbf{a}\|},\ \mathbf{y} = \frac{\mathbf{b}}{\|\mathbf{b}\|} \text{ by definition above} \\
&\Rightarrow \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{a}\|\|\mathbf{b}\|} \leq 1 && \text{\# scalar multiplication rule: } \langle c_1\mathbf{v}_1, c_2\mathbf{v}_2 \rangle = c_1 c_2 \langle \mathbf{v}_1, \mathbf{v}_2 \rangle \\
&\Rightarrow \langle \mathbf{a}, \mathbf{b} \rangle \leq \|\mathbf{a}\|\|\mathbf{b}\| && \text{\# multiply both sides by } \|\mathbf{a}\|\|\mathbf{b}\|
\end{aligned}
$$

If we replace $\mathbf{a}$ by $-\mathbf{a}$ we see by Equation (2) that $\|-\mathbf{a}\| = \|\mathbf{a}\|$. Further, by the scalar multiplication rule for dot products with $c_1 = -1$ and $c_2 = 1$ we have $\langle -\mathbf{a}, \mathbf{b} \rangle = -\langle \mathbf{a}, \mathbf{b} \rangle$. Taken together these results imply that $-\langle \mathbf{a}, \mathbf{b} \rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|$.

Next note that the absolute value of x, $|x|$, is defined to be

$$
|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0. \end{cases} \tag{6}
$$

So one way to see the Cauchy–Schwarz inequality is by setting x equal to $\langle \mathbf{a}, \mathbf{b} \rangle$ in Equation (6). Then when $\langle \mathbf{a}, \mathbf{b} \rangle \geq 0$ we have $\left|\langle \mathbf{a}, \mathbf{b} \rangle\right| = \langle \mathbf{a}, \mathbf{b} \rangle$ and we showed above that $\langle \mathbf{a}, \mathbf{b} \rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|$. Similarily, when $\langle \mathbf{a}, \mathbf{b} \rangle < 0$ we have $\left|\langle \mathbf{a}, \mathbf{b} \rangle\right| = -\langle \mathbf{a}, \mathbf{b} \rangle$ and we showed that $-\langle \mathbf{a}, \mathbf{b} \rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|$. Together these give us $\left|\langle \mathbf{a}, \mathbf{b} \rangle\right| \leq \|\mathbf{a}\|\|\mathbf{b}\|$, that is, the Cauchy–Schwarz inequality. $\square$

3.1 Derivatives

We will also denote the derivative of a vector $\mathbf{r}$ with respect to t by $\dot{\mathbf{r}}$, $\dfrac{d\mathbf{r}}{dt}$ or $\mathbf{r}'(t)$. Note also that by definition

$$\frac{d\mathbf{r}}{dt} := \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

4 Arc Length

Consider a segment of a parametric curve $\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}} + k(t)\,\hat{\mathbf{k}}$ between two points $P = \mathbf{r}(t)$ and $Q = \mathbf{r}(t + \Delta t)$, as shown in Figure 4.

The next question we might ask is what is the length of the segment of the curve between P and Q? This length is called the arc length and is denoted by Δs and can be approximated by the chord length $\|\Delta \mathbf{r}\|$.

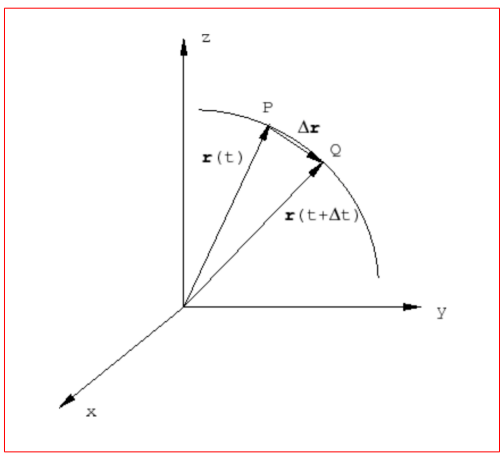

Figure 4: $\Delta \mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$

Specifically we can see from Figure 4 that

$$\|\Delta r\| = \|\mathbf{r}(t + \Delta t) - \mathbf{r}(t)\|$$

and so

$$\Delta s \approx \|\mathbf{r}(t + \Delta t) - \mathbf{r}(t)\|$$

Now we can compute the infinitesimal arc length ds:

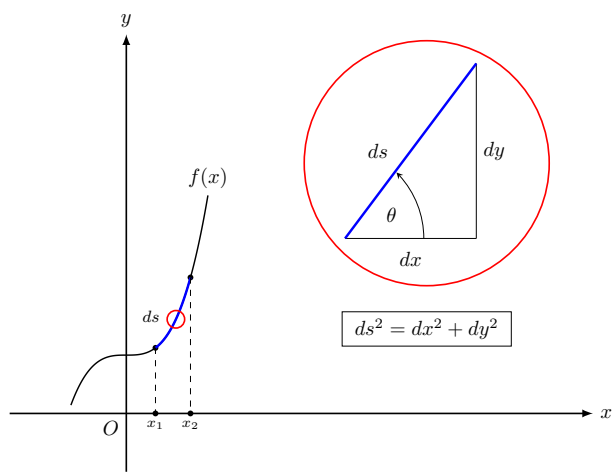

Figure 5: $f(x)$, ds and the Pythagorean Theorem

$$
\begin{aligned}
\Delta s &\approx \|\Delta \mathbf{r}\| && \text{\# approximate } \Delta s \text{ by } \|\Delta \mathbf{r}\| \\
&= \|\mathbf{r}(t+\Delta t) - \mathbf{r}(t)\| && \text{\# } \Delta \mathbf{r} = \mathbf{r}(t+\Delta t) - \mathbf{r}(t) \\
&= \|\mathbf{r}(t+\Delta t) - \mathbf{r}(t)\| \cdot \frac{\Delta t}{\Delta t} && \text{\# multiply by } 1 = \frac{\Delta t}{\Delta t} \\
&= \left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t && \text{\# assume } \Delta t > 0 \\
\Rightarrow\ & \Delta s \approx \left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t && \text{\# combine expressions for } \Delta s \\
\Rightarrow\ & \lim_{\Delta t \to 0} \Delta s = \lim_{\Delta t \to 0} \left[\left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t \right] && \text{\# take the limit of both sides} \\
\Rightarrow\ & ds = \lim_{\Delta t \to 0} \left[\left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t \right] && \text{\# } \Delta t \to 0 \Rightarrow \Delta s \to 0 \Rightarrow \lim_{\Delta t \to 0} \Delta s = ds \\
\Rightarrow\ & ds = \lim_{\Delta t \to 0} \left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \lim_{\Delta t \to 0} \Delta t && \text{\# product rule for limits [1]} \\
\Rightarrow\ & ds = \left\| \lim_{\Delta t \to 0} \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \lim_{\Delta t \to 0} \Delta t && \text{\# for a normed vector space } X \ \lim \|x_n\| = \|\lim x_n\| \\
\Rightarrow\ & ds = \left\| \frac{d\mathbf{r}}{dt} \right\| \cdot dt && \text{\# } \lim_{\Delta t \to 0} \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{d\mathbf{r}}{dt},\ \lim_{\Delta t \to 0} \Delta t = dt
\end{aligned}
$$

Interestingly we can also see this using the Pythagorean theorem. Specifically, if we move an infinitesimal distance in the x and y directions we get a triangle in the x-y plane who's sides have length dx and dy and who's hypotenuse is ds. This scenario is shown for some curve $f(x)$ in Figure 5.

In the two dimensional case where the parameterization our $x = g(t)$ and $y = h(t)$ we have $\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}}$. Further, since derivative is a linear operator [10] we know that

$$\frac{d\mathbf{r}}{dt} = \frac{dg}{dt}\,\hat{\mathbf{i}} + \frac{dh}{dt}\,\hat{\mathbf{j}} \tag{7}$$

The Pythagorean theorem tells us that $ds = \sqrt{dx^2 + dy^2}$. Using this expression for ds we can see that

$$\begin{aligned}
ds &= \sqrt{dx^2 + dy^2} && \text{\# Figure 5} \\
&= \sqrt{dg^2 + dh^2} && \text{\# switch to parametric form: } x = g(t) \text{ and } y = h(t) \\
&= \sqrt{dg^2 + dh^2} \cdot \frac{dt}{dt} && \text{\# multiply by } 1 = \frac{dt}{dt},\ dt > 0 \\
&= \sqrt{dg^2 + dh^2} \cdot \left[\sqrt{\frac{1}{dt^2}} \cdot dt\right] && \text{\# } \frac{x}{x} = \sqrt{\left(\frac{1}{x}\right)^2} \cdot x = \sqrt{\frac{1}{x^2}} \cdot x,\ x > 0 \\
&= \sqrt{\left(dg^2 + dh^2\right) \cdot \left(\frac{1}{dt^2}\right)} \cdot dt && \text{\# simplify} \\
&= \sqrt{\left(\frac{dg}{dt}\right)^2 + \left(\frac{dh}{dt}\right)^2} \cdot dt && \text{\# simplify} \\
&= \sqrt{g'(t)^2 + h'(t)^2} \cdot dt && \text{\# switch notation from Leibniz} \rightarrow \text{Lagrange: } \frac{df}{dt} = f'(t) \\
&= \left\|\frac{d\mathbf{r}}{dt}\right\| \cdot dt && \text{\# Equation (7) and the definition of } \|\cdot\|
\end{aligned}$$

Now we can see that

$$\frac{ds}{dt} = \left\|\frac{d\mathbf{r}}{dt}\right\|$$

and so

$$\frac{ds}{dt} = \|\dot{\mathbf{r}}\| \tag{8}$$

4.1 The Unit Tangent Vector T

Since $\frac{ds}{dt} = \left\|\frac{d\mathbf{r}}{dt}\right\| = \|\dot{\mathbf{r}}\|$ we also know that the unit tangent vector $\mathbf{T} = \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} = \frac{d\mathbf{r}}{ds}$ [6]. One way to see this is to notice that

$$
\begin{aligned}
\mathbf{T} &= \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} && \text{\# definition of the unit tangent vector } \mathbf{T} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\|\dot{\mathbf{r}}\|} && \text{\# } \dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt} \text{ (definition of } \dot{\mathbf{r}}\text{)} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\left(\frac{ds}{dt}\right)} && \text{\# } \|\dot{\mathbf{r}}\| = \frac{ds}{dt} \text{ (Equation (8))} \\
&= \frac{d\mathbf{r}}{ds} && \text{\# simplify}
\end{aligned}
$$

5 The Line Integral

Consider the definite integral shown in Figure 6. Here we want to find the area S that is above the line segment $[a, b]$ and below the curve $f(x)$.

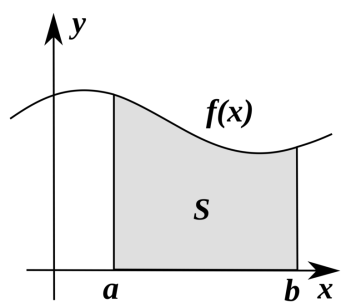

Figure 6: Definite Integral: The area $\mathbf{S} = \int_a^b f(x)\,dx$

The line integral, denoted $\int_C f(x, y)\,ds$, is similar except that here we want to find the area above the curve C and below the function $f(x, y)$ (so the line integral is by definition in three dimensions). The comparison between the definite integral $\int_a^b f(x)\,dx$ and the line integral $\int_C f(x, y)\,ds$ is shown in Figure 7.

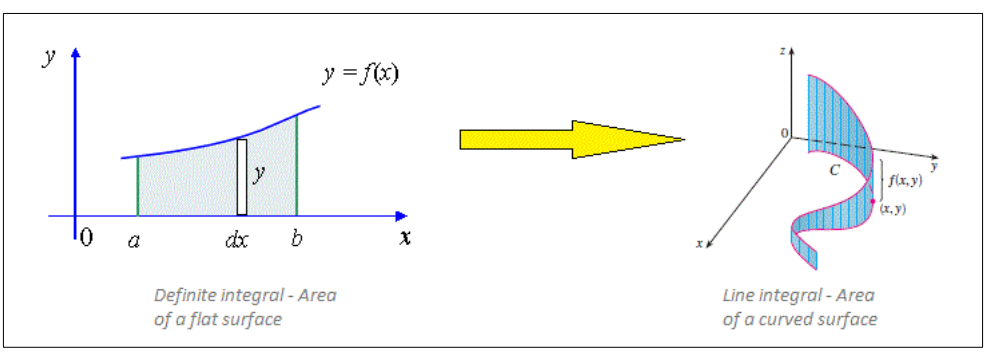

Figure 7: Definite vs. Line Integrals

5.0.1 Parameterizing the Curve C

The first step in solving a line integral is to find a parameterization for the curve C. We would like to find a parameterization such that the integral reduces to an integral over a single variable t. For example, suppose our curve C is a circle of radius r where we want to integrate over the part of the circle in the first quadrant. That is, our curve C is $x^2 + y^2 = r^2$ and so the parametric form of x and y would be $x(t) = g(t) = r\cos(t)$ and $y(t) = h(t) = r\sin(t)$.

We also need to specify the limits of integration for our parameterization. Since the curve C is the circle of radius r in the first quadrant, the limits of integration are $a = 0$ and $b = \frac{\pi}{2}$ (that is, $t \in [0, \frac{\pi}{2}]$). Putting this all together we get

$$\int_C f(x,y)\,ds = \int_0^{\frac{\pi}{2}} f(h(t), g(t))\,ds = \int_0^{\frac{\pi}{2}} f(r\cos(t), r\sin(t))\,ds$$

Ok, but what is ds? As we saw above ds is the infinitesimal arc length and that the Pythagorean theorem tells us that $ds = \sqrt{dx^2 + dy^2}$ (Figure 5). Using this expression for ds we can see that

$$\begin{aligned}
ds &= \sqrt{dx^2 + dy^2} && \text{\# by the Pythagorean theorem} \\
&= \sqrt{dg^2 + dh^2} && \text{\# switch to parametric form: } x = g(t) \text{ and } y = h(t) \\
&= \sqrt{dg^2 + dh^2} \cdot \frac{dt}{dt} && \text{\# multiply by } 1 = \frac{dt}{dt},\ dt > 0 \\
&= \sqrt{dg^2 + dh^2} \cdot \left[\sqrt{\frac{1}{dt^2}} \cdot dt\right] && \text{\#} \ \frac{x}{x} = \sqrt{\left(\frac{1}{x}\right)^2} \cdot x = \sqrt{\frac{1}{x^2}} \cdot x,\ x > 0 \\
&= \sqrt{\left(dg^2 + dh^2\right) \cdot \left(\frac{1}{dt^2}\right)} \cdot dt && \text{\# simplify} \\
&= \sqrt{\left(\frac{dg}{dt}\right)^2 + \left(\frac{dh}{dt}\right)^2} \cdot dt && \text{\# simplify} \\
&= \sqrt{g'(t)^2 + h'(t)^2} \cdot dt && \text{\# switch notation from Leibniz} \rightarrow \text{Lagrange: } \frac{df}{dt} = f'(t)
\end{aligned}$$

Now we can rewrite our line integral in terms of a single variable t as follows:

$$\int_C f(x,y)\,ds = \int_a^b f(g(t), h(t))\sqrt{g'(t)^2 + h'(t)^2} \cdot dt$$

At this point our line integral has been reduced a definite integral over one variable (which hopefully we can solve).

5.1 Vector Fields

In vector calculus and physics, a vector field is an assignment of a vector to each point in a subset of some space [5]. For example, a vector field in the plane can be visualized as a collection of arrows with a given magnitude and direction, each attached to a point in the plane. Vector fields are often used to model the speed anddirection of a moving fluid throughout space, or the strength and direction of some force, such as the magnetic or gravitational force, as it changes from one point to another point.

For example, the wind velocity vector field for the 2011 Joplin, MO tornado [9] is shown in Figure 8. Here the color represents the wind speed $\|\mathbf{v}\|$, where $\mathbf{v}$ is the wind velocity vector.

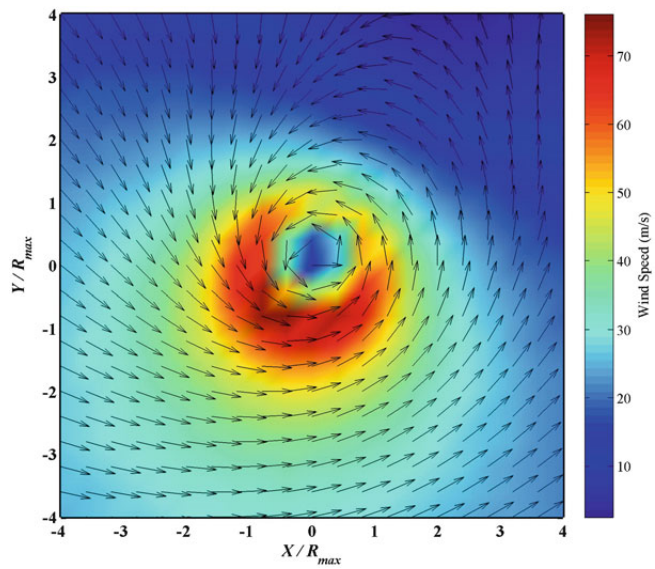

Figure 8: 2011 Joplin, MO Tornado Wind Velocity Vector Field

The general form of a vector field (here in three dimensions) is

$$\vec{F}(x, y, z) = P(x, y, z)\,\hat{\imath} + Q(x, y, z)\,\hat{\jmath} + R(x, y, z)\,\hat{k}$$

where P, Q, and R are scalar functions.

5.2 Line Integrals of Vector Fields

5.3 Fundamental Theorem for Line Integrals

5.4 Conservative Vector Fields

5.5 Green's Theorem

5.6 Stoke's Theorem

6 Surface Integrals

7 Conclusions

8 Acknowledgements

Thanks to Dave Neary who pointed out that using the reverse triangle inequality in my proof of the Cauchy–Schwarz inequality was overkill. In particular, $\|\mathbf{x}-\mathbf{y}\| \geq 0$ for all $\mathbf{x}$ and $\mathbf{y}$ by Equation (2), so the triangle inequality (or reverse triangle inequality) was not needed.

9 Appendix A

I had thought that one way to prove the Cauchy–Schwarz inequality is to use the reverse triangle inequality, but Dave Neary pointed out that I didn't need it. So I'm leaving the derivation of the reverse triangle inequality in this appendix.

Theorem 9.1. Reverse Triangle Inequality: $\big|\|\mathbf{x}\| - \|\mathbf{y}\|\big| \leq \|\mathbf{x}-\mathbf{y}\|$

We can derive the reverse triangle inequality from the triangle inequality [7, 16] by observing that

$$
\begin{aligned}
\|\mathbf{x}\| &= \|\mathbf{x} + (-\mathbf{y}+\mathbf{y})\| && \text{\# add 0 to } \mathbf{x}\text{: } \mathbf{x} = \mathbf{x}+0 = \mathbf{x}+(-\mathbf{y}+\mathbf{y}) \\
&= \|(\mathbf{x}-\mathbf{y})+\mathbf{y}\| && \text{\# addition is associative: } \mathbf{v}_1+(\mathbf{v}_2+\mathbf{v}_3) = (\mathbf{v}_1+\mathbf{v}_2)+\mathbf{v}_3 \\
&\leq \|\mathbf{x}-\mathbf{y}\| + \|\mathbf{y}\| && \text{\# by the triangle inequality: } \|\mathbf{v}_1+\mathbf{v}_2\| \leq \|\mathbf{v}_1\| + \|\mathbf{v}_2\| \\
&\Rightarrow \|\mathbf{x}\| \leq \|\mathbf{x}-\mathbf{y}\| + \|\mathbf{y}\| && \text{\# combine expressions for } \|\mathbf{x}\| \\
&\Rightarrow \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x}-\mathbf{y}\| && \text{\# subtract } \|\mathbf{y}\| \text{ from both sides}
\end{aligned}
$$

and

$$
\begin{aligned}
\|\mathbf{y}\| &= \|\mathbf{y} + (-\mathbf{x}+\mathbf{x})\| && \text{\# add 0 to } \mathbf{y}\text{: } \mathbf{y} = \mathbf{y}+0 = \mathbf{y}+(-\mathbf{x}+\mathbf{x}) \\
&= \|(\mathbf{y}-\mathbf{x})+\mathbf{x}\| && \text{\# addition is associative} \\
&\leq \|\mathbf{y}-\mathbf{x}\| + \|\mathbf{x}\| && \text{\# by the triangle inequality} \\
&\Rightarrow \|\mathbf{y}\| \leq \|\mathbf{y}-\mathbf{x}\| + \|\mathbf{x}\| && \text{\# combine expressions for } \|\mathbf{y}\| \\
&\Rightarrow \|\mathbf{y}\| - \|\mathbf{x}\| \leq \|\mathbf{y}-\mathbf{x}\| && \text{\# subtract } \|\mathbf{x}\| \text{ from both sides} \\
&\Rightarrow -\Big[\|\mathbf{y}\| - \|\mathbf{x}\|\Big] \geq -\|\mathbf{y}-\mathbf{x}\| && \text{\# multiply both sides by -1: } -1*(\mathbf{v}_1 \leq \mathbf{v}_2) \Rightarrow -\mathbf{v}_1 \geq -\mathbf{v}_2 \\
&\Rightarrow \|\mathbf{x}\| - \|\mathbf{y}\| \geq -\|\mathbf{x}-\mathbf{y}\| && \text{\# } \|\mathbf{y}-\mathbf{x}\| = \|-(\mathbf{y}-\mathbf{x})\| = \|\mathbf{x}-\mathbf{y}\| \text{ by Equation (2)} \\
&\Rightarrow -\|\mathbf{x}-\mathbf{y}\| \leq \|\mathbf{x}\| - \|\mathbf{y}\| && \text{\# rearrange}
\end{aligned}
$$

Combining the expressions for $\|\mathbf{x}\| - \|\mathbf{y}\|$ we get $-\|\mathbf{x}-\mathbf{y}\| \leq \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x}-\mathbf{y}\|$, that is, the reverse triangle inequality $\big|\|\mathbf{x}\| - \|\mathbf{y}\|\big| \leq \|\mathbf{x}-\mathbf{y}\|$. $\square$

LaTeX

`https://www.overleaf.com/read/fgtfvmgdkbhh`

References

[1] Alex Svirin. Properties of Limits. https://math24.net/properties-limits.html. [Online; accessed 14-November-2021].

[2] Dan Margalit and Joseph Rabinoff. Cofactor Expansion. https://textbooks.math.gatech.edu/ila/determinants-cofactors.html, May 2020. [Online; accessed 18-March-2022].

[3] David Meyer. Notes on Change of Basis. https://davidmeyer.github.io/qc/change_of_basis.pdf, January 2019. [Online; accessed 14-March-2022].

[4] Frank Jones. Euclidean Space. http://www.owlnet.rice.edu/~fjones/chap1.pdf, Aug 2004. [Online; accessed 03-November-2021].

[5] Galbis, A. and Maestre, M. *Vector Analysis Versus Vector Calculus*. Universitext. Springer New York, 2012.

[6] Gregory Hartman. Unit Tangent and Normal Vectors. https://sites.und.edu/timothy.prescott/apex/web/apex.Ch12.S4.html, Nov 2021. [Online; accessed 08-March-2022].

[7] Jean H. Gallier. Vector Norms and Matrix Norms. https://www.cis.upenn.edu/~cis515/cis515-11-sl4.pdf, 2020. [Online; accessed 10-March-2022].

[8] Math Open Reference. Proof of the Law of Cosines. https://www.mathopenref.com/lawofcosinesproof.html, 2011. [Online; accessed 16-March-2022].

[9] NIST. Joplin Missouri Tornado 2011. https://www.nist.gov/disaster-failure-studies/joplin-tornado-ncst-investigation, Aug 2011. [Online; accessed 10-March-2022].

[10] Wikipedia Contributors. Linearity of Differentiation — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Linearity_of_differentiation&oldid=1026177205, 2021. [Online; accessed 1-September-2021].

[11] Wikipedia Contributors. Rule of Sarrus — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Rule_of_Sarrus&oldid=1050637323, 2021. [Online; accessed 18-March-2022].

[12] Wikipedia Contributors. Cross product — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Cross_product&oldid=1070828862, 2022. [Online; accessed 18-March-2022].

[13] Wikipedia Contributors. Dot Product — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Dot_product&oldid=1074690360, 2022. [Online; accessed 17-March-2022].

[14] Wikipedia Contributors. Jacobi identity — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Jacobi_identity&oldid=1075896608, 2022. [Online; accessed 18-March-2022].

[15] Wikipedia Contributors. Right-Hand Rule — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Right-hand_rule&oldid=1069678571, 2022. [Online; accessed 18-March-2022].

[16] Wikipedia Contributors. Triangle Inequality — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Triangle_inequality&oldid=1074965160, 2022. [Online; accessed 10-March-2022].